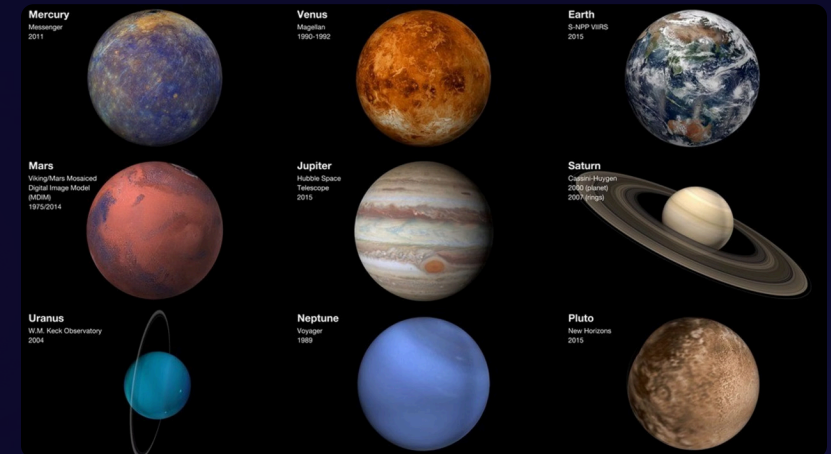




Universe Expansion + possible ends

Solar System

- **8 planets orbiting the Sun:**
- **Beyond planets:**
 - Vibrant tapestry of moons, asteroids, and comets.
 - These celestial bodies provide clues to the solar system's formation and evolution.
- **The Planets:**
 - **Mercury:** Scorching, rocky surface; closest to the Sun.
 - **Venus:** Thick, toxic atmosphere; hottest planet.
 - **Earth:** Supports life; has liquid water and a protective atmosphere.
 - **Mars:** Known as the "Red Planet"; may harbor evidence of past water.
 - **Jupiter:** Largest planet; a majestic gas giant with over 80 moons.
 - **Saturn:** Famous for its stunning ring system; second-largest gas giant.
 - **Uranus:** Ice giant; rotates on its side with a pale blue-green color.
 - **Neptune:** Deep blue ice giant; has supersonic winds and faint rings.



The Expanding Universe

The universe is not static; it has been continuously expanding since its origin in the Big Bang. This groundbreaking discovery was made by Edwin Hubble in **1929**, who observed that distant galaxies are systematically moving away from us, a phenomenon known as Hubble's Law.

Evidence of Expansion

Hubble's observations showed that galaxies are receding from Earth, and the further away they are, the faster they move. This provides direct evidence for the expanding fabric of spacetime itself.

Accelerating Expansion

In the late 1990s, observations of distant supernovae revealed an even more astonishing fact: the universe's expansion is not slowing down due to gravity, but is actually accelerating, driven by a force called dark energy.

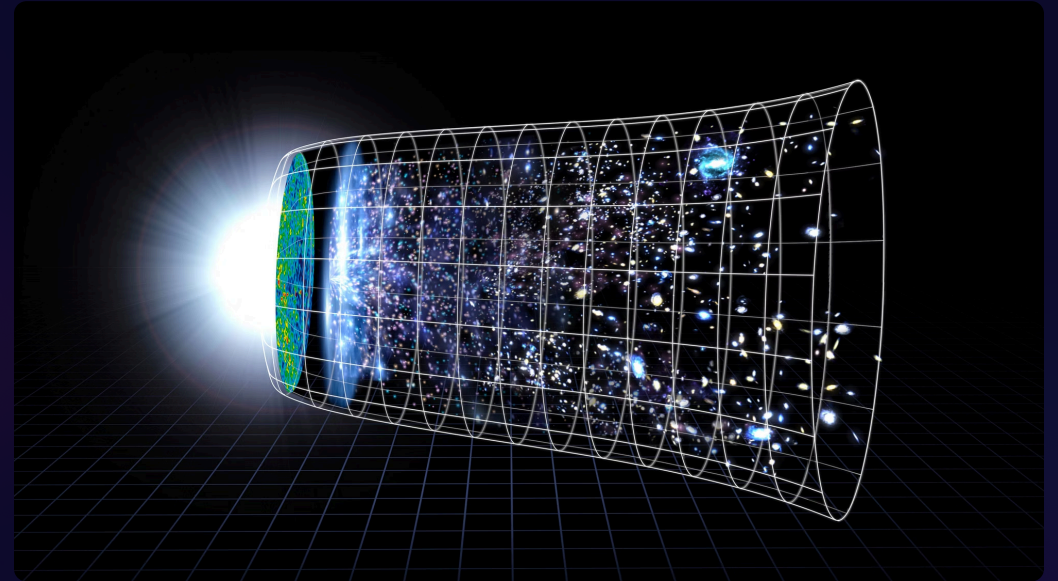
Edwin Hubble

- Edwin Hubble's pioneering work at Mount Wilson Observatory (1920s) revolutionized our understanding of the universe.
- Key discoveries:
 - "Spiral nebulae" were actually distant galaxies beyond the Milky Way.
 - Observed Cepheid variable stars in these galaxies to measure distances:
 - Pulsation period directly relates to intrinsic luminosity ("standard candles" for cosmic distances).
 - Found that distant galaxies are systematically moving away:
 - The farther a galaxy is, the faster it recedes.
 - Led to Hubble's Law: recession velocity proportional to distance, with H_0 as Hubble's constant
- Impact of Hubble's work:
 - Shifted the view of the cosmos from static to dynamic and expanding.
 - Laid the foundation for the Big Bang theory.
- Honors:
 - The Hubble Space Telescope (launched in 1990) was named after him for his contributions.



Dark Energy

- Late 1990s: Observations of distant supernovae revealed surprising findings:
 - The universe's expansion is accelerating, not slowing down due to gravity.
- This led to the proposal of dark energy:
 - A mysterious force counteracting gravity and driving accelerated expansion.
- Dark energy:
 - Constitutes ~68% of the universe's total energy density.
 - Drives the ever-faster expansion of the universe.
- Its true nature remains one of modern cosmology's greatest unsolved mysteries.
- Inspires intense research and new theories about the fundamental makeup of the cosmos.



How Do We Study Universe Expansion?

Astronomers employ several ingenious methods to study the universe's ongoing expansion:



Observing Redshift

Analyzing the shift of light towards the red end of the spectrum from distant galaxies, indicating they are moving away from us and the universe is expanding.



Type Ia Supernovae

Utilizing these extremely bright and consistent explosions as 'standard candles' to measure cosmic distances and the rate of expansion.



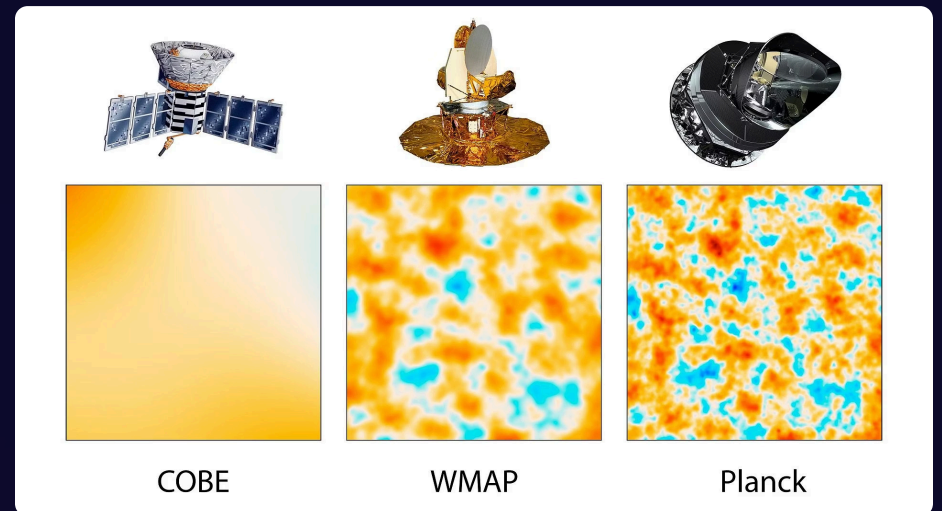
Cosmic Microwave Background

Studying the faint afterglow radiation from the Big Bang, which provides crucial evidence about the early universe's expansion and structure.

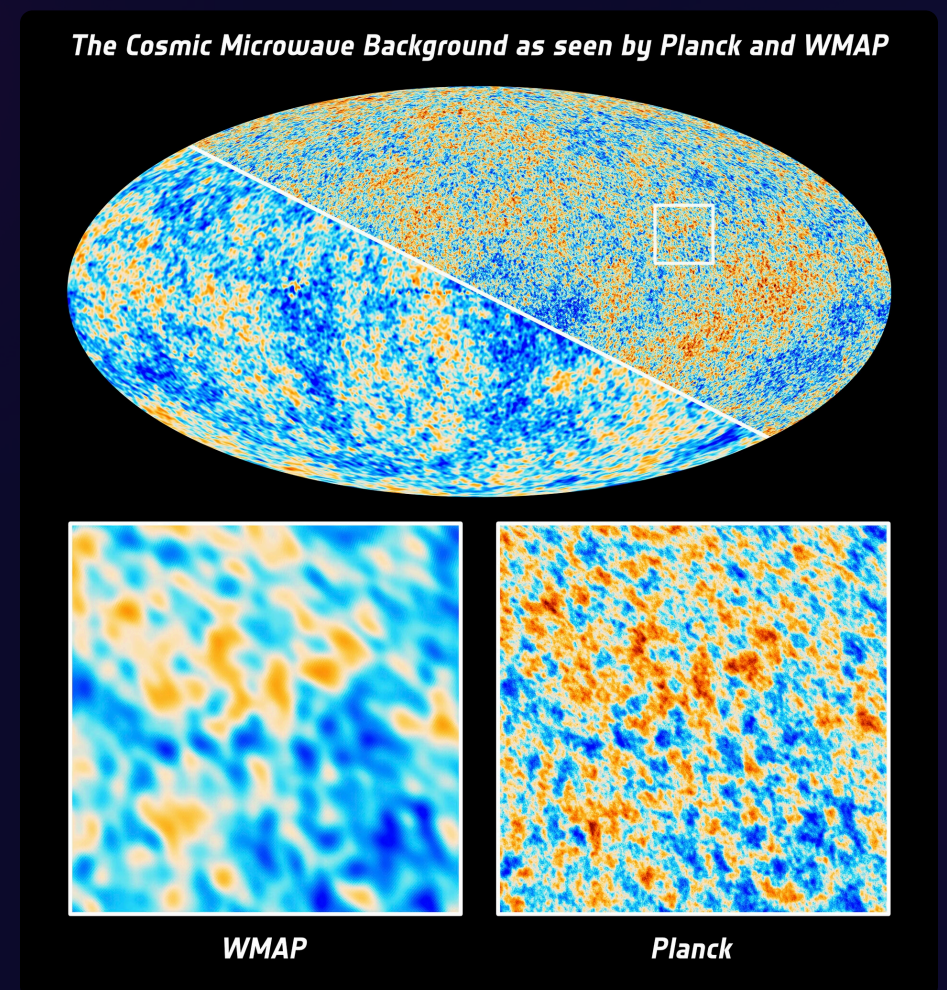
The Cosmic Microwave Background

Early missions like COBE, followed by WMAP and Planck, have progressively mapped the CMB in greater detail, providing invaluable data on the early universe.

- The Cosmic Microwave Background (CMB):
 - Oldest light in the universe, a faint radiation glow permeating all of space.
 - Origin: Afterglow from when the universe became transparent (~380,000 years after the Big Bang).
 - Formed when electrons and protons combined to create neutral hydrogen, allowing photons to travel freely.
- Discovery:
 - Accidentally detected in 1965 by Arno Penzias and Robert Wilson.
 - Initially identified as persistent, uniform noise in their radio antenna.
 - Later confirmed as the CMB, a relic of the hot, dense early universe.
- Mapping the CMB:
 - Provided invaluable data on the early universe.
- Insights from the CMB:
 - Tiny temperature fluctuations reveal seeds of galaxy formation.
 - Denser regions eventually collapsed under gravity to form cosmic structures.
 - Critical data on the universe's age, composition, and expansion rate.
- Importance:
 - A cornerstone of modern cosmology, offering a precise picture of the early universe.



Evolution in our understanding and mapping capabilities of the Cosmic Microwave Background (CMB), showing progressively higher resolution and detail of the faint temperature fluctuations.



Subtle variations in the CMB, crucial for refining cosmological models.

Possible Fates of the Universe

Scientists propose several dramatic scenarios for the ultimate destiny of our universe, each dependent on factors like its total energy density and expansion rate:

The Big Freeze (Heat Death)

The universe continues to expand indefinitely, eventually running out of all available energy, leading to a cold, dark, and empty void as stars burn out and black holes evaporate.

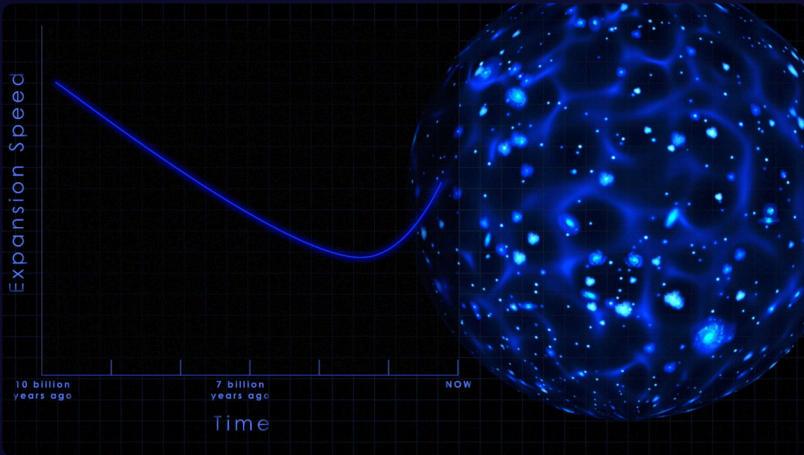
The Big Crunch (Gravitational Collapse)

If the universe's density is high enough, gravity could eventually halt its expansion and pull all matter back together into an infinitely hot, dense singularity, reversing the Big Bang.

The Big Rip (Accelerating Expansion)

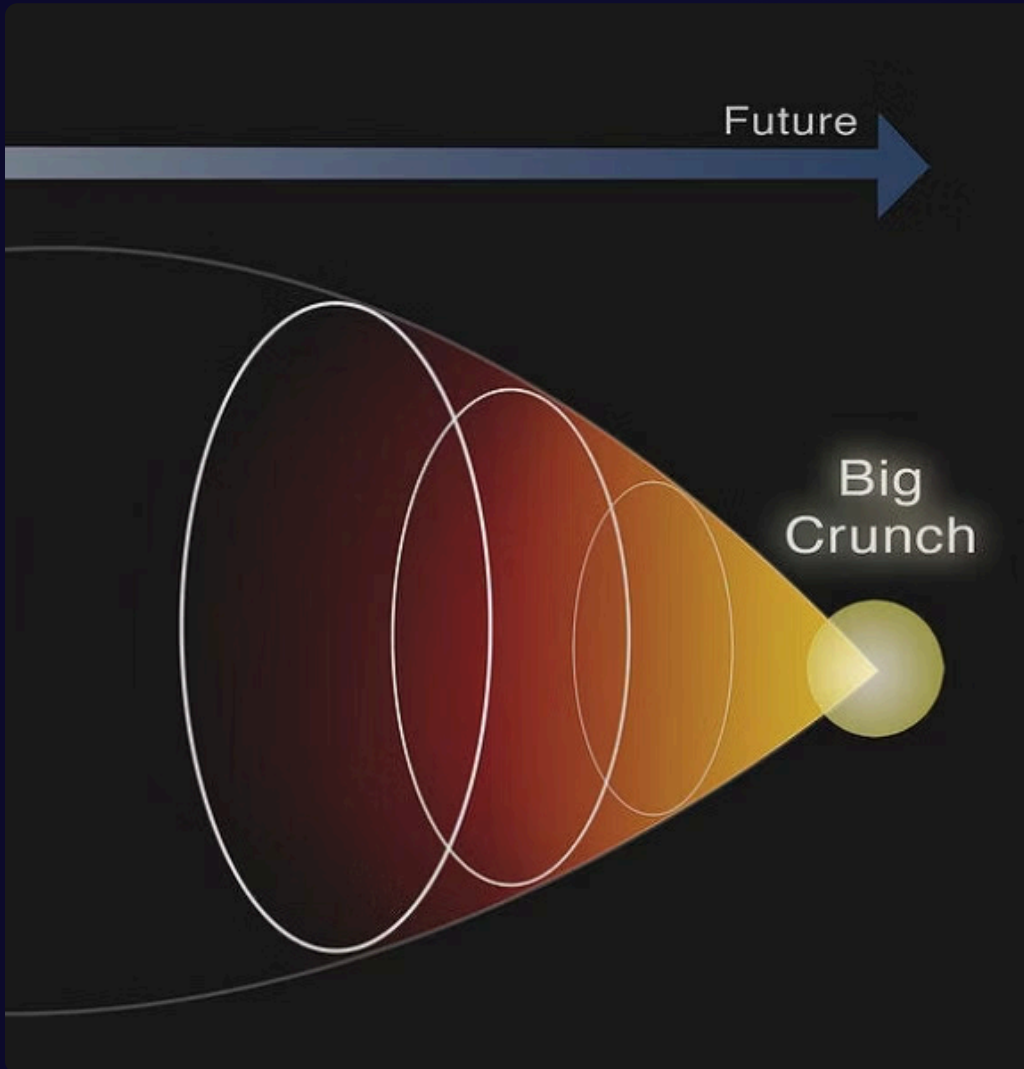
Driven by dark energy, the universe's accelerating expansion could eventually become so powerful that it tears apart galaxies, stars, planets, and even atoms themselves.

The Big Freeze: A Cold, Dark Future



- **The Big Freeze (Heat Death):**
 - Describes a future where the universe expands indefinitely.
 - Leads to a state of maximum entropy, with energy uniformly distributed.
 - No further work or processes can occur.
- **Over time:**
 - Stars burn out, exhausting their fuel.
 - Black holes evaporate via Hawking radiation.
 - Results in a cold, dark, and empty void.
- **Current understanding:**
 - Accelerating expansion suggests the Big Freeze is the most likely fate of the universe.

The Big Crunch: Gravitational Collapse



- **The Big Crunch:**
 - A scenario where gravity overcomes expansion if the universe contains enough matter and dark energy weakens.
 - Results in a reverse timeline of the Big Bang, leading to a cosmic collapse.
- **What happens in this scenario:**
 - Galaxies merge into super-galaxies.
 - Stars collide more frequently.
 - Temperatures across the universe rise dramatically.
 - All matter and energy compress into an infinitely hot, dense singularity.
- **Current understanding:**
 - Less likely due to the accelerating expansion driven by dark energy.
 - Remains theoretically possible under different cosmological conditions.

The Big Rip: Accelerating Expansion

- **The Big Rip:**

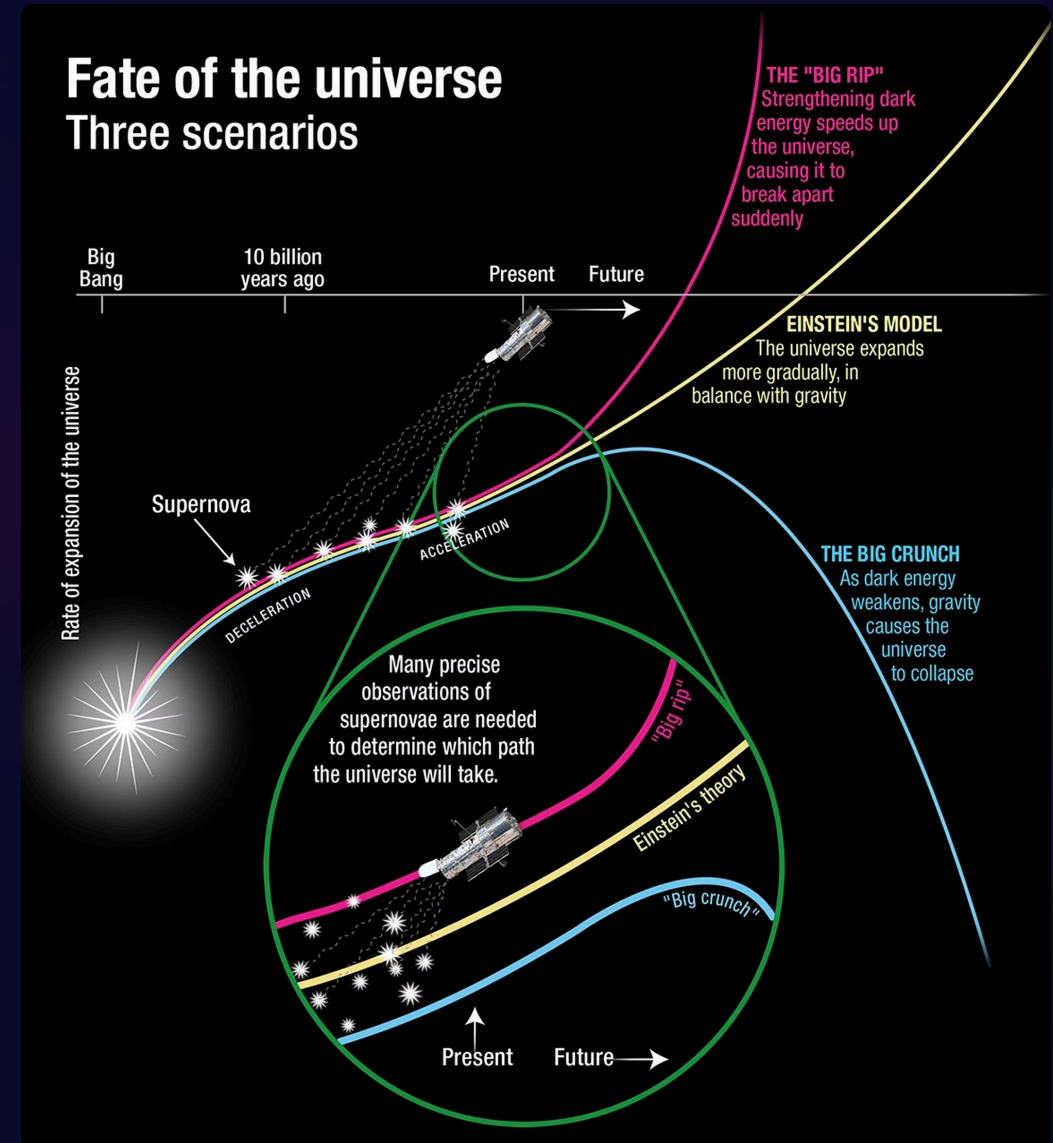
- A scenario where dark energy strengthens, causing the universe's expansion to accelerate uncontrollably.
- Expansion overcomes all fundamental forces.

- **Timeline of events:**

- Galaxies separate beyond observable limits.
- Solar systems are torn apart.
- Planets are ripped from their stars.
- Even atoms are destroyed as space expands faster than the speed of light locally.

- **Predicted occurrence:**

- Could happen in ~22 billion years if dark energy behaves as "phantom energy."
- Phantom energy has an equation of state more negative than -1.



Timeline to the End

The universe's ultimate fate, often referred to as the Big Freeze or Heat Death, unfolds over timescales that are truly immense and difficult to comprehend. From the cessation of star formation to the evaporation of the last black holes, the cosmos will gradually descend into a cold, dark, and empty void.

1

Star Formation Ends

In approximately **100 trillion years**, the supply of gas and dust required for new star formation will be exhausted. The universe will become a much darker place.

2

Last Stars Die

Between **100 trillion and 1 quadrillion years** from now, the remaining stars, mostly red dwarfs, will burn through their fuel and slowly fade away, leaving behind stellar remnants.

3

Black Hole Evaporation

Over periods far exceeding the lifetime of the last stars, black holes will gradually evaporate via Hawking radiation. This process will continue for incredibly vast timescales, eventually leaving only elementary particles.

